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Acronyms

Abbreviation	Meaning
AI	Artificial Intelligence
CONOPS	Concept of Operations
Dx	Deliverable X
DB	Data Base
DMA	Differential Mobility Analyzer
DPO	Data Protection Officer
DR	Detection Rate
EU	European Union
EWS	Early-Warning System
FAR	False-Alarm Rate
GDPR	General Data Protection Regulation
KPI	Key-Performance Indicator
LEA	Law-Enforcement Agency
ML	Machine-Learning
MLOps	Machine-Learning Operations
M	Month
MS	Mass Spectrometer
RMS	Remote Monitoring Station
SEN	SENSITIVE
SIT	Substance Intelligence Tool
T	Task
TCO	Total Cost of Ownership
TRL	Technology-Readiness Level

TTR	Time-To-Result
WP	Work Package
WMS	Warehouse Management Systems
UC	Use-Case
UX	User Experience
FR	Functional requirement (e.g., <i>"system must do this"</i>)
NFR	Non-Functional Requirement (e.g., <i>"system must be secure"</i>)
V&V	Verification and Validation

Executive Summary

This document specifies the EU PENTION solution using a rigorous, software systems engineering requirements engineering approach. PENTION is framed as an assistive system for Law-Enforcement in which high-flow physical sampling and analysis are assisted by AI components and other intelligence. This specification consolidates requirements elicited directly (via workshops and focus groups) and indirectly (surveys) from PENTION stakeholders, and structures them without prescribing an implementation. What is more, the analysis is informed by a pilot study of related EU initiatives—ARMADILLO (GHB), NARCOSIS (NPS), ARIEN (AI for trafficking), BorderSens (sensor/nano/data), PARSEC & IFlows (non-intrusive screening), and MUTRATE (holistic readiness)—via card-sorting to extract reusable requirements and patterns. As a result of our requirements engineering exercise, the PENTION system shall comprise two complementary modalities: PENTION-S (static) for parcel/warehouse screening and PENTION-M (mobile) for open-area screening and checkpoints. High-flow vapour sampling feeds a DMA-MS detector. The Situational-Intelligence Tool (SIT) based on Artificial Intelligence performs peak deconvolution, interference filtering, and classification for drugs, precursors, and by-products. Outputs are automatic, explainable, chain-of-evidence-ready and integrated with RMS for prioritisation. A secured Machine-Learning Operations (MLOps) loop keeps models current as backgrounds and concealment tactics evolve against an open, updateable substance/precursor database.

1. Introduction

1.1 Purpose

The purpose of this document is to prepare a full specification for the system solution intended in the EU PENTION action. The intended target audience for this document covers all stakeholders of the EU PENTION action as well as any relevant policymaker interested in the workings of the solution and the preliminaries required to specify it. To this purpose, the document contains an outline of the scope and contextual foundations behind the requirements engineering campaign reported in this document. Subsequently, the document elaborates on said requirements and on the architecture reflected by them.

1.2 Scope and Materials

The scope for the contents of this document is limited to specifying, analysing, and reporting the requirements expressed directly, mainly via workshops and focus group, and indirectly mainly through large-scale surveys, by the EU PENTION stakeholders. It also describes the intended architecture derived from these requirements, without expressing any deterministic representation of the EU PENTION solution or its implementation, which are addressed in the corresponding tasks of the project.

The document, rather, is intended as a source of key requirements for PENTION systems designed to support forensic investigation of Law-Enforcement Agencies (LEAs) either for the scenarios intended in PENTION or those foreseen for related projects of the future. The reference materials behind this document encompass also an investigation of key pieces of literature reflecting the material in the requirements, including any material stemming from any connected projects, primarily:

- ARMADILLO → Detection of gamma-hydroxybutyrate (GHB) substances;
- NARCOSIS → Rapid detection of new psychoactive substances (NPSs);
- ARIEN → AI-driven tools to monitor and analyse drug trafficking in real-time;
- BorderSens → Sensor technologies, nanotechnology and data analysis to detect a wide range of drugs;
- PARSEC & IFlows → non-intrusive screening technologies to detection and identification of illicit goods, materials and substances;
- MUTRATE → Holistic Readiness Level;
- METEOR → Holistic Readiness Level for revolutionized cargo screening;

The projects contents, either freely available online or linked on their respective websites, shall be investigated with a simple card-sorting approach to identify and categorize all material that might prove useful to refine the intended targets of this document.

1.3 Context Overview

PENTION is, by conceptual definition, a security-grade system that combine physical sampling hardware with edge analytics and cloud-assisted intelligence to detect and disrupt synthetic-drug production and trafficking across EU operational contexts.

On the physical layer, PENTION-S (static) is designed for integration with parcel hubs/warehouses, while PENTION-M (mobile, vehicle-mounted) enables open-area sweeps and checkpoint screening.

High-flow vapour sampling feeds a DMA-MS analyser whose spectra are interpreted by the SIT AI for peak deconvolution, interference filtering, and classification of drugs, precursors, and degradation products.

At the cyber layer, results are generated automatically in an explainable manner, logged with chain-of-evidence, and integrated with Risk Management Systems (RMS) for prioritisation. A secure MLOps loop updates models as backgrounds, concealment techniques, and substances evolve, backed by an open, updateable threat database.

Cross-cutting concerns include non-intrusive screening, privacy, safety, cybersecurity, and compliance with EU priorities (low FAR, high DR, sub-minute analysis, TRL-7 validation). The system is going to be validated in realistic pilots (e.g., Limburg region) and designed for 24/7, high-throughput operations and swift adoption by LEAs/Customs.

1.4 Research Methods

Reader's Digest. PENTION requirements were obtained through mixed-methods and validated through (a) formal modelling (for correctness) and (b) triangulation between RIEC and Spanish Guardia Civil (for completeness). Thematic coding, requirements reverse elicitation, MoScow categorization and other state-of-the-art approaches were used for analysis and synthesis.

This methodology section operationalises PENTION's mission—detect and disrupt synthetic drugs—into a governed and testable system lifecycle. The following provides a crisp, battle-tested requirements engineering methodology drawing on established practices in requirements engineering [2] and tailored to the PENTION concept and KPIs (sub-minute analysis, $\geq$ TRL 7). The approach is end-to-end, from framing to field pilots, with concrete artifacts and gates followed to derive, validate, triangulate, model, and analyse requirements for synthesis and automated reasoning.

On the one hand, the PENTION's mission starts by establishing governance (G0) and stakeholder intent (G1), then grounding the work in real operational scenarios and a CONOPS (G2). Requirements are elicited and structured with explicit fit criteria and verification

methods (G3), while hazards, security, privacy, and ethics are engineered as first-class requirements (G4). The conceptual architecture is refined into allocatable components, interfaces, and budgets (G5), and every item remains traceable from policy need to field KPIs via a live trace matrix and an Alloy consistency layer.

On the other hand, data and AI are engineered as subsystems (G6), including governed datasets, measurable model quality (per-substance metrics, explainability, robustness), and an MLOps loop for safe updates. Verification & validation (G7) proceeds from environmental tests, and field pilots with statistically powered acceptance criteria for timing. Implementation gates (G8–G10) progressively integrate and harden the system across its cyber and physical boundaries, and pilots deliver TRL-7 evidence (G11) in target contexts (e.g., Limburg in NL and parcel hubs). The result is a non-intrusive, explainable, evidence-ready PENTION system, backed by an auditable chain of design-time and run-time evidence to support operational adoption by LEAs and Customs across the EU.

From an analytic perspective, elicited requirements underwent Prioritization following the established MoSCoW [3] approach followed by auto-tagging of requirements to Use-Cases & KPIs. Finally, to verify the compliance, consistency, correctness, and completeness of requirements, the analysis campaign featured the *Alloy* [4] requirements engineering and modelling approach, as outlined by Mannering et al. [5]. Alloy is a lightweight, declarative language for requirements of modeling and analysis based on relational logic. You describe the world with signatures (types such as Sample, Analyzer, Result), relations between them, and constraints. Hard constraints live in facts (always true), scenario assumptions in predicates (can be invoked for exploration), and properties to verify in assertions. The Alloy Analyzer compiles the model to satisfiability conditions and then searches for all finite instances within a chosen scope to either generate examples that satisfy your constraints (run) or find counterexamples to assertions (check). This makes Alloy ideal for exposing ambiguity, contradictions, and hidden edge cases early—well before code or hardware. Applying Alloy to PENTION, we turned key requirements into executable logic as follows:

- Structure & roles: model PENTION-S/M samplers, DMA-MS analyzer, SIT AI, Threat DB, RMS, operators, and evidence artifacts as signatures and relations.
- Non-negotiables as facts: automatic outputs and explainability; chain-of-evidence for any alarm; an open/updateable Threat DB with ≥ 6 substances; non-intrusive screening; and timing guards (e.g., sampling $\leq 1\text{--}3$ minutes by scenario, analysis readiness).
- KPIs as assertions: encode $DR \geq 0.90$ (semi-closed), $DR \geq 0.80$ (open), and $FAR < 5\%$ over sets of Event; count true positives/false alarms with integer arithmetic and check across bounded worlds.

- Scenarios as predicates: a *SmallWorld* harness ensures both open and semi-closed cases exist, and items are screened, letting the Analyzer produce concrete witness instances stakeholders can inspect.
- What-if exploration: relax or tighten constraints to see when the model becomes unsatisfiable (over-constrained) or admits pathological instances (under-specified); use the Analyzer's unsatisfiable core to pinpoint conflicting constraints.
- Behavior: Add a state signature with next relations to model samples → analyze → report traces and constrain temporal properties over finite steps.

Integration into the requirements V&V loop: map each SHALL to a fact/assertion/predicate (tagged with the requirement ID); run checks in CI to guard against regressions as the specification evolves; and turn counterexamples into test cases and MBSE refinements. The outcome is a continuously analyzable, unambiguous PENTION baseline where policy goals—KPIs, legal/ethics, safety/security—are enforced as logic rather than opinion.

2. Requirements Validity and Description

Reader's Digest. A considerable proportion the PENTION stakeholder consortium participated in the requirements activities. All participants were highly experienced practitioners and collectively prioritised our target to scenario 1 (from the proposal) as target scenario, while also providing valuable perspectives across each considered scenario.

The elicitation involved the entire consortium. This section is aimed at representing such a sample using any descriptive statistics and outlines available from the elicitation mechanisms adopted according to details provided in the previous section.

First, the figure below outlines the types of stakeholders involved in our survey, therefore outlining the representativeness per sector; a total of 15 highly expert individuals participated to our survey, which was kept opened for a total of 50 days before analysis was kicked off, for a response rate of over 67%.

The distribution per class of said individuals reflects the valuable experience of each practitioner; everyone has a background of 20+ years of involvement in the themes and contexts afferent to PENTION IA and therefore reflects a considerable amount of base knowledge.

The next section outlines the system overview obtained by modelling the requirements elicited via all data sources available to this campaign and analyzed through the Alloy requirements modelling and analysis tool. The Alloy specification along with the consolidated list of requirements is available upon request, while a recap sample of the functional and non-functional requirements is reported in Sec. 3.5.

3. Results Overview and Key Requirements

The requirements elicited using the methodology described above were subsequently modeled to assess their correctness and completeness. As previously stated, the Alloy formal specification language and its assert-based completion modelling approach were used to attain a correct and complete specification; at the same time a narrative modelling approach [8] and open coding [9] were employed to express the requirements and overall system view in a structured and narrative form. The result of this process is summarized in the tables below (functional requirements) and further on (non-functional requirements).

3.1 Functional Requirements

Who will use the system and where
• Parcel hub screeners at warehouses and ports need to check parcels and pallets without opening them, while keeping lines moving. • Road and border teams need to check vehicles quickly and decide “clear” or “inspect more”. • Patrol teams drive through neighborhoods and industrial areas to find hidden drug labs and mark likely spots on a map. • Event security teams want to scan tents and gathering areas in places like festivals or concerts. • Investigation and risk staff need to sort alerts, direct teams, and track how much useful information the system adds. • Evidence officers and prosecutors need a reliable record of what was found, where, when, and by whom, that stands in court. • System maintainers, trainers, and buyers need a sturdy, easy-to-service tool with clear screens, short training, and good value for money.
What the system must do
• Screen parcels, pallets, and containers without opening them; give a clear answer fast; keep queues short. • Check vehicles at the roadside or at borders; make quick “clear” or “inspect more” decisions. • Sweep open areas to locate hidden labs; turn measurements into map “hotspots” so teams know where to go next. Sample upwind and downwind of suspected source to isolate target fugitive emission and identification of contrabands. • Work in crowded places, such as festivals, to find risk areas without stopping people or opening bags. • Recognize many substances and ingredients, including new ones, and expand its list quickly when new threats appear. • Explain its results in simple words (why it raised an alert); avoid screens that only experts can read. • Send results to dispatch so teams can be directed and send decisions back to warehouse systems so parcels can keep moving.

• Record every step (who, what, where, when) in a tamper-proof log for possible use in court.
How it should perform
• Be fast. Parcel answers should come in about a minute; road checks and spot checks should finish in a few minutes. • Find most real threats and raise a few mistaken alerts (aim higher indoors than outdoors; keep mistaken alerts to only a few in a hundred checks). • Handle high volume in parcel hubs and keep up with busy shifts. • Work all day, every day in heat, cold, rain, and vibration; no fragile supplies or special gases. • Be simple to use. Short training; clear “what to do next”; large, readable screens.
Data, learning, and updates
• Keep a living library of substances and ingredients; add new entries within days of confirmation by a partner lab. • Adapt to each region as it learns local background conditions, without exposing personal data. • Show why an alert happened, in plain words, and with a short highlight of the signal that led to it.

3.2 Non-Functional Requirements

Safety, privacy, and trust
• Do not open parcels and do not touch people; collect only what is needed from the air around the item or area. • Store the least personal data possible; control who can access what; keep a full audit trail. • For serious alerts, support a full chain of handling so the record can be used in court; allow second-method confirmation at a partner forensic lab when needed.
Connections and day-to-day running
• Connect to warehouse systems to fetch shipment details and return pass/hold decisions. • Connect to dispatch so tasks reach the right teams at the right time. • Share warning signals with national and EU groups that track new substances. • Provide clear dashboards: how many checks were done, how many alerts were real, coverage of routes, and overall benefit versus effort.

3.3 Scenario-Based Requirements Representation

To summarize the requirements in context, the flow diagram below illustrates the operational logic behind PENTION systems. The diagram is represented as a simple box-and-line representation of how PENTION output is intended to be used in action and shall be read from left to right.

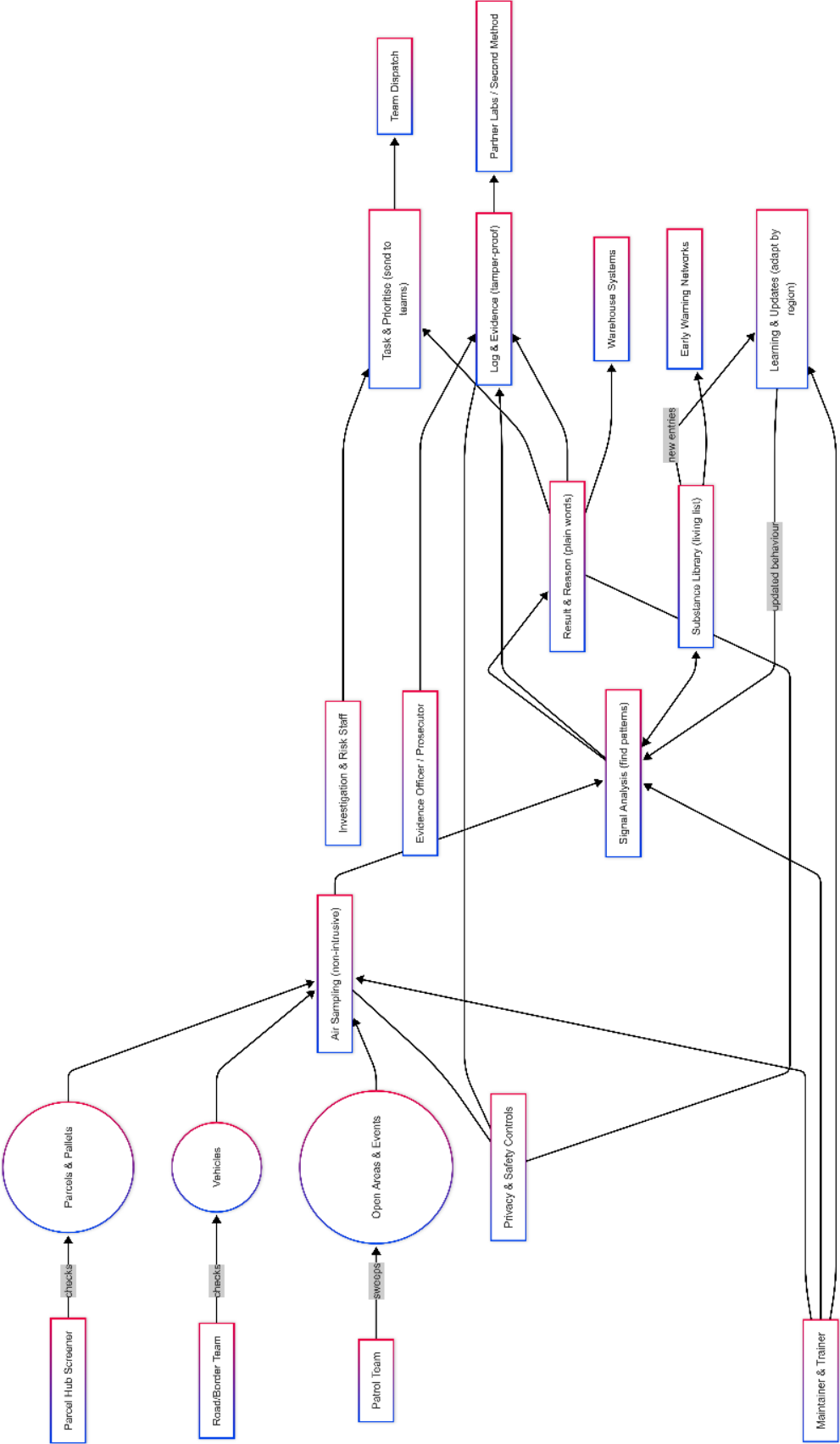


Figure 1 - overview of the PENTION dataflow.

How to read the diagram:

- On the left are **people**; at the top are the **places and items** they check.
- The Centre shows the **screening pipeline**: collect air around the item or area → analyze the signal → produce a result with a short reason → send tasks or decisions.
- Every action is **logged for evidence**. A **living library** of substances supports analysis and gets **updated** as new threats appear.
- On the right are the **connections** to daily operations and partner groups.
- **Privacy and safety** wrap up key steps, so the system stays non-intrusive, respectful of rights, and ready for court if needed.

Realistic Usage Scenario Example

Morning at the parcel hub

At 07:30, a **parcel hub screener** begins a shift at a busy warehouse. Parcels and pallets roll past the station (**Parcels & Pallets**). The screener docks the first pallet under the hood. The system pulls a small amount of air from around the packages (**Air Sampling**, non-intrusive) and immediately checks the signal (**Signal Analysis**). A short message appears: “**Possible match to an ingredient often used to make party pills**,” with a plain-language reason and a short signal highlight (**Result & Reason**). The system records who did what and when (**Log & Evidence**) and sends a “hold” note back to the warehouse screen for this pallet (**Warehouse Systems**). Because the match is stronger than a simple warning, the system also passes the case to the team that sorts and directs field work (**Task & Prioritize → Team Dispatch**).

All along, **Privacy & Safety Controls** ensure that no parcel is opened, and no personal details are stored. The screener sees only what they need to clear or hold the pallet.

Mid-morning: risk triage and a quick route change

In the operations room, **investigation and risk staff** see the new alert on their dashboard. They check nearby alerts and decide to send the **patrol team** to sample air around several blocks near the industrial zone and the outdoor market (**Task & Prioritise → Team Dispatch**; people → **Open Areas & Events**). The staff also see that vehicles from a small courier are passing through the same area; they inform the **road and border team** to be ready at a roadside stop.

Late morning: the street sweep and a vehicle stop

The **patrol team** drives a loop around the industrial streets (**Open Areas & Events → Air Sampling**). As the van rolls, the system gathers air and tags each sample with place and time. The signal checker turns raw measurements into a simple map with warmer and cooler spots

(**Signal Analysis** → **Result & Reason**). A spot near a lock-up unit light up. The system records all steps (**Log & Evidence**) and sends a high-priority task to the nearest road team (**Task & Prioritize** → **Team Dispatch**).

A **road team** stops a small van (**Vehicles**). A quick air check around the bodywork and vents find a less certain signal (**Air/particle Sampling** → **Signal Analysis** → **Result & Reason**). The message reads: “**Unclear—possible trace of the same ingredient as earlier. Recommend a closer look.**” The team calls for secondary action. The system logs every decision and who made it (**Log & Evidence**).

Noon: second-method confirmation and legal handover

Given the link between the pallet, the hotspot, and the vehicle, the duty **evidence officer** opens a case. The system bundles the records—who, what, where, when, and the reasons shown to each operator—and packages them for handover (**Log & Evidence**). A small sample from the van air filter is sent to a **partner forensic lab** for a second method check (**Log & Evidence** → **Partner Labs / Second Method**). The officer can see the complete chain of handling: timestamps, people, labels, and the exact messages shown. Nothing personal is exposed that isn’t strictly needed, and access is controlled by role (**Privacy & Safety Controls** around logging and results).

Early afternoon: new knowledge, shared safely

The partner lab confirms a **new variant** of a known ingredient. The **substance library** adds a new entry and notes how its signal looks in this region (**Substance Library**). The system learns from this update so that future checks in the same region spot it more clearly (**Substance Library** → **Learning & Updates** → **Signal Analysis**). A short notice is also shared with national groups that collect warnings about new drugs (**Substance Library** → **Early Warning Networks**). The message uses plain language and includes no personal details.

Mid-afternoon: keeping the warehouse moving

Back at the hub, the earlier pallet is still on hold. With the new library entry, the system re-checks the stored air sample from that pallet (**Signal Analysis** with **Learning & Updates**). This time the match is confident or otherwise validated. The system sends a clear “hold and isolate” instruction to the warehouse screen and prints a short, plain note for the screener: “**Confirmed: parcel group contains an ingredient often used to make party pills.**” (**Result & Reason** → **Warehouse Systems**). The log updates the case with this new decision (**Log & Evidence**).

Late afternoon: a check-up and training moment

A **maintainer** runs a short service routine—filter change, self-check—and confirms everything passes (**Maintainer & Trainer → Air/Particle Sampling, Signal Analysis**). They also push the new library entry and a short job aid that explains the change in plain words (**Maintainer & Trainer → Learning & Updates**). A **trainer** visits the road team to show them how today's messages looked and how to act on “unclear” results with minimal delay (**Maintainer & Trainer → Learning & Updates → people**).

Evening: event perimeter

At a music festival, **event security** working with the **patrol team** circles outer tents (**Open Areas & Events**). The system gathers air gently and never touches people or bags (**Air Sampling with Privacy & Safety Controls**). A mild warning appears near a vendor tent; the message explains the likely cause and suggests a brief revisit in 15 minutes as the crowd thins (**Result & Reason**). The revisit returns clear, so no action is taken; the system records both checks (**Log & Evidence**). The festival continues without disruption; alternatively, security should be alerted if positive.

Next day: from story to practice

At the daily review, **investigation and risk staff** see that the combined parcel-street-vehicle pattern raised useful alerts with only a few mistaken ones. The dashboard shows average time-to-answer, checks per hour, and how much extra useful information the system added to the day's decisions (**Result & Reason, Task & Prioritize, Log & Evidence**). A short note goes to the **programme manager** about the gains versus effort and a suggestion to extend the street loop by two blocks.

3.4 User Classes and Characteristics

This section identifies the various user classes that we anticipate will use this product and describes their pertinent characteristics. Some requirements might pertain only to certain user classes. Identify the favored user classes. User classes represent a subset of the stakeholders described in the vision and scope of documentation. User class descriptions are a reusable resource. The reader shall find relevant user class descriptions here, starting from a high-level overview of the PENTION technical space, recapped in the figure below.

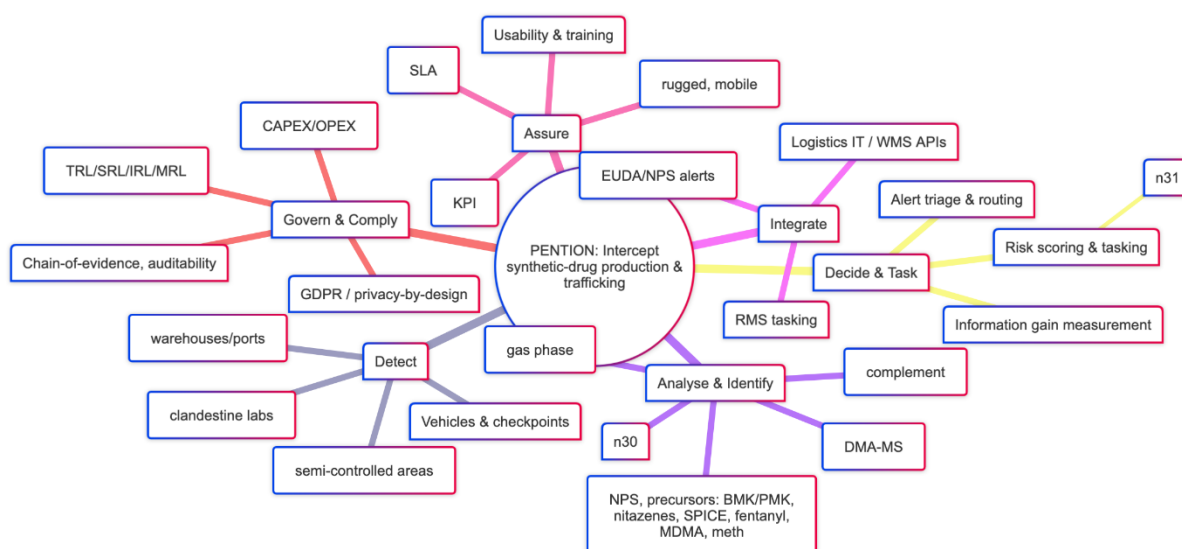


Figure 2 - an overview of the PENTION technical space; bounded boxes represent concepts and players in the space, relations represent their association, while colours reflect a different domain/type of association.

The technical space recaps the key technical features intended in PENTION and outlines their hierarchical relations. For example, PENTION technical support is designed towards deciding and tasking around: (a) risk scoring and tasking; (b) alert triaging and routing; (c) information gain measurement. The goal model recaps therefore the following conceptual areas:

Categories observed: (1) Functional–Detection/Screening, (2) Interface (Logistics/RMS/EWS), (3) AI/Data, (4) NFR (Performance/Throughput/Robustness/Usability), (5) Security/Privacy, (6) Legal/Compliance, (7) Governance/Deployment—each covered by components and goals.

Use-contexts: (1) parcels/ports/warehouses, (2) vehicles/checkpoints, (3) open-area lab sweeps, (4) semi-controlled events.

Substances/precursors (from Substances sheet and prior inputs out of related projects): SPICE, nitazenes, fentanyl, MDMA, methamphetamine; BMK/PMK and related glycidate; residue (swab) as complementary mode.

Compliance & evidence: GDPR, auditability, chain-of-evidence; lab MS corroboration where required.

Operations: theatre-specific retraining; info-gain metrics; cost/benefit and readiness gates.

In addition, the underlying use-cases and actors intended in PENTION, as reflected by our requirements elicitation campaign, is reported further below following a classical pattern of Use-case maps (actors → scenarios).

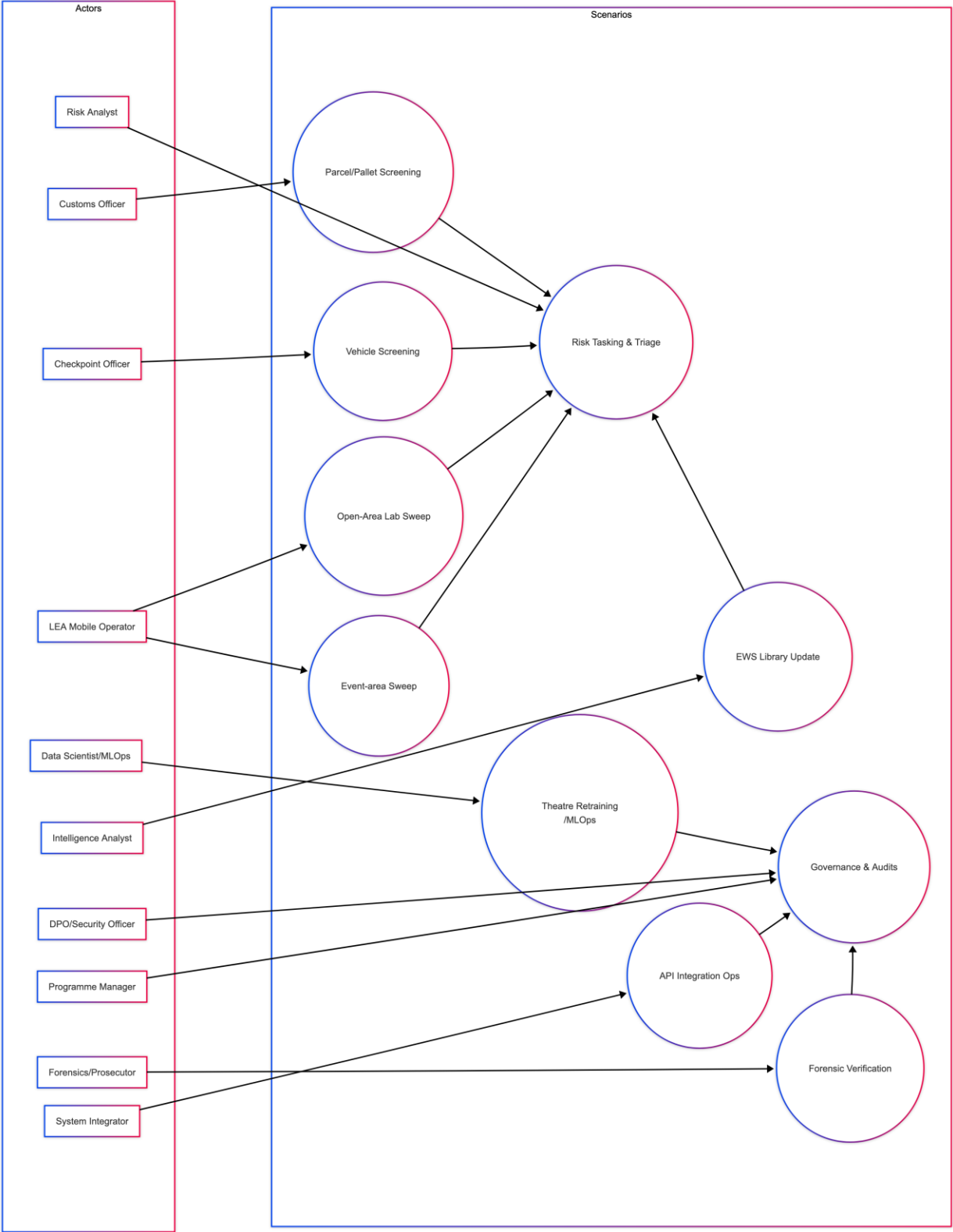


Figure 3 - an overview of the PENTION use-cases; on the left-hand side appear the intended and recovered PENTION actors while on the right-hand side appear all potential use-cases for the PENTION system.

3.5 System Actors Specification

The actors presented below are extracted directly from the requirements and scenario-based elicitation exercises enacted in the PENTION requirements engineering campaign. Here below follows the entire list of elicited actors, arranged per category:

Primary operational users

- Customs Officer (Border/Port/Parcel)
 - Goals: fast and reliable screening of parcels/pallets/containers; low FAR; clear alarms.
 - Environment: warehouses, postal hubs, ports, border checkpoints.
 - Frequency: high, shift based.
 - Characteristics: trained operator, time pressure, logistics-aware; cares about throughput & false alarms.
- LEA Patrol / Mobile Sampling Operator
 - Goals: open-area/vehicle sweeps for clandestine labs, route coverage; on-van analytics.
 - Environment: roads, rural/industrial zones, parking areas, event perimeters.
 - Frequency: patrol cycles; incident driven.
 - Characteristics: rugged equipment, quick TTR; weather & vibration tolerance.
- Checkpoint Officer (Road/Border)
 - Goals: non-intrusive vehicle screening; quick clear/hold decisions.
 - Environment: roadside checkpoints, fixed/mobile sites.
 - Characteristics: short dwell time per vehicle; needs concise, auditable results.

Investigation, analysis & decision

- Risk Analyst / Tasking Officer
 - Goals: fuse risk indicators, task mobile/static assets; triage alerts; measure information gain.
 - Characteristics: dashboards, scoring rules, watchlists, explainability.
- Intelligence Analyst (EWS)
 - Goals: monitor NPS trends & alerts; maintain substance watch lists; share signals.
 - Characteristics: interoperable feeds; provenance; quick library updates.
- Prosecutor / Forensics Officer
 - Goals: chain-of-evidence, admissible results, cross-method corroboration.
 - Characteristics: audit trails, versioned models/libraries, independent verification paths.

Technical delivery & operations

- Data Scientist / MLOps Engineer
 - Goals: theatre-specific retraining, drift detection, model registries, explainability.
 - Characteristics: reproducible pipelines, benchmark suites, information-gain metrics.
- System Integrator (Logistics/RMS/EWS)
 - Goals: robust APIs with WMS/parcel IT, RMS tasking, EWS updates.
 - Characteristics: standards-based schemas, retries/backoff, SLA monitoring.
- Field Engineer / Maintenance
 - Goals: calibration, consumables, uptime; ruggedization checks.
 - Characteristics: in-field serviceability; no specialist gases/liquid N₂.
- UX Lead / Trainer
 - Goals: low-burden workflows; operator training content; human factors.
 - Characteristics: role-based UIs; alerts clarity; accessibility.

Governance, policy & oversight

- Security Officer / DPO
 - Goals: GDPR compliance, data minimization, access control, audit.
- Program Manager / Buyer
 - Goals: CAPEX/OPEX vs. detection gains; TCO.
- EU Policy / Courts (indirect)
 - Goals: lawful operation; clear standards of evidence; regulatory alignment.

3.6 Technical/Technological Requirements

PENTION-S sampler

The PENTION-S sampler is being designed to provide reliable and consistent air sampling, operating with a stable airflow between 5 and 450 L/min. Its future high-precision flowmeter will ensure accurate monitoring of the sampling flow and precise calculation of the total air volume collected. The unit will be powered by 230 V AC, 50 Hz, allowing seamless integration into fixed installations once deployed.

In terms of construction, PENTION-S will feature a robust, sealed IP54 enclosure and a durable suction hose over 10 meters long, engineered to withstand demanding mechanical conditions. A non-return valve will ensure unidirectional airflow and prevent contamination during use. Designed for challenging environments, the sampler will operate reliably between –25 °C and 50 °C and at 0–80% relative humidity, ensuring consistent performance across a wide range of future operating conditions.

PENTION-M sampler

The PENTION-M unit is planned to incorporate a set of technical features that ensure reliable, autonomous, and contamination-free air sampling. It will support remote operation, allowing the sampler to be controlled or monitored without direct interaction, which increases safety and flexibility in the field. The device will operate with a stable airflow between 5 and 450 L/min, measured with a high-precision flowmeter to guarantee accurate calculation of the sampled air volume. Although conceived as a static unit, it will be powered by one or multiple batteries or portable generator, resulting in at least 5 hours of autonomy, ensuring uninterrupted operation even in environments where access to external power is limited. Additionally, the system will include GNSS system, providing traceability and accurate geolocation for every sampling event.

From a construction standpoint, PENTION-M will use lightweight yet highly resistant materials, offering thermal and electrical insulation to protect internal components. A non-return valve will ensure unidirectional flow and prevent contamination of both the device and the sampling environment. The enclosure will meet IP54 protection, enabling the unit to withstand rain, wind, hail, and snow, making it suitable for outdoor conditions. Finally, it will be engineered to operate reliably across a wide temperature range from -25°C to 50°C and under 0–80% relative humidity, ensuring consistent performance regardless of environmental conditions.

PENTION portable system

The PENTION transportable system will be designed to integrate the DMA/MS unit into a mobile platform while meeting several key technical requirements. The system, with an expected total weight of around 330 kg, will need a structurally reinforced mounting solution to ensure stability and protect sensitive analytical components. It will operate from a 230 V AC, 50 Hz power supply (internal generators), requiring a future electrical installation capable of delivering 14 A continuously plus an additional 3 A intermittently for the nitrogen generator. To guarantee analytical integrity during operation, all nitrogen lines will be built using clean Teflon tubing free of contamination. The system will also incorporate a dedicated exhaust hose that will exhaust gases safely to the outside of the vehicle, avoiding recirculation or contamination inside the cabin.

From a thermal, acoustic, and mechanical standpoint, the unit will require engineered solutions to ensure stable performance. A tailored heat-management system will be implemented to extract and dissipate the significant thermal load generated by the analyser, pumps, and electronics. The internal environment will be thermostat to ensure adequate temperature and humidity inside the van. To control noise, expected to reach high levels due to the roughing pump, an acoustic padded insulation system will be included to reduce sound transmission and maintain operator comfort. Additionally, the installation will feature a mechanical shock absorber, damping system designed to keep reduce vibrations transmitted

to the DMA/MS system below 5 m/s², ensuring both hardware protection and stable analytical performance.

PENTION list of substances

A collaboration between technical experts and end-users was established to define a list of potential substances of interest for the PENTION project. Specifically, this activity was coordinated by MION, and NP, MIRPN, ITI, RIEC and PPC were involved in the definition of the list of substances. The process for defining the list of PENTION substances was based on individual proposals for the different partners, including the reasons and the impacts of the substances, specifically, the proposals were:

- MIRPN proposed a list of 11 precursors
- NP, in collaboration with RIEC, proposed a list with 4 drugs and 2 main precursors
- ITI proposed several documents with chemical information and potential impact of different substances, including several drugs and many precursors

After a deep analysis of the different alternatives and a review of some literature, MION proposed a list of substances for PENTION containing 4 drugs and 2 main precursors, the final agreed list is the following:

- Amphetamine (drug)
- MDMA (drug)
- Cocaine (drug)
- Methamphetamine (drug)
- PMK (precursor)
- BMK (precursor)

Including the glycidyl derivatives.

3.7 Operational Requirements

PENTION-S sampler

For PENTION-S, the operational requirements are centered on enabling fast and reliable sampling within a fixed installation. The system will work with a dedicated sampling cabin designed to allow quick and easy loading and unloading of parcels, ensuring smooth operation in environments where many packages are handled. To simplify the process for the operator, the device will offer predefined sampling profiles adapted to different parcel types, ensuring consistent and efficient sampling regardless of the item being inspected.

Each sampling cycle will be completed in less than one minute, minimizing delays and allowing high throughput. Additionally, the unit will incorporate an automatic cleaning and anti-contamination system, ensuring that the cabin and the sampling circuit remain free from residues between tests. This reduces the risk of cross-contamination and guarantees that each sample collected is clean, reliable, and representative.

PENTION-M sampler

For PENTION-M, the operational requirements focus on enabling quick and efficient field use by a single operator. The device is being designed so that one operator can handle all deployment steps, from transport to operation, without additional assistance. If the final weight is below 5 kg, it will be carried manually; if it falls between 5 and 9 kg, it will be transported using a small trolley, ensuring safe and ergonomic handling in all scenarios. The system will also support fast deployment, allowing the operator to set it up and begin sampling within seconds, which is essential in dynamic or time-sensitive environments.

During each sampling event, the device will automatically record its GNSS position, providing precise geolocation data for traceability and documentation. The operational workflow is further optimized by ensuring that each sampling cycle can be completed in less than three minutes, enabling rapid assessments in the field and allowing the operator to cover multiple locations efficiently. Overall, these requirements make PENTION-M a portable, quick-to-use, and highly practical tool for mobile air-sampling operations.

PENTION portable system

For the PENTION transportable system, the operational requirements focus on ensuring that both the DMA/MS equipment and the operator can function safely and efficiently inside the vehicle. During operation, the interior cabin will need to maintain a controlled environment, with a temperature between 15 and 30 °C and humidity levels between 20% and 80%, since the analytical system depends on stable climatic conditions to perform accurately. Because all measurements will be carried out with the vehicle stationary, the shock mount system must be capable of keeping transmitted vibrations below 5 m/s², protecting sensitive components and ensuring reliable analytical results.

In addition to supporting the equipment's needs, the system must also accommodate the operator's working conditions. The vehicle will include a dedicated operator area, separate from the equipment zone, allowing tasks to be carried out comfortably and safely. This operator space will be equipped with its own climate-control system to maintain optimal temperature and humidity independently of the heat generated by the DMA/MS unit. Furthermore, an acoustic-insulation system will be incorporated to create a physical and acoustic barrier between both areas, reducing exposure to noise from pumps and fans and ensuring a quieter and more ergonomic working environment.

3.8 Requirements Full Elicited List

For a full list of the functional requirements and substances discovered in the reported requirements engineering campaign, please refer to the tables in Appendix.

4. Requirements Traceability Matrix

The user classes contained in this document, as well as the requirements and any modelled data reported here, were derived from repeated cues across our available data and requirement sets derived from it (e.g., descriptions on warehouse/logistics operators dominate hub scenarios; customs/border and vehicle inspection teams appear in port/road checks; LEA/forensics appear in clandestine lab and chain-of-evidence workflows; AI/MLOps and Integrators recur around library updates, retraining, and IT links; DPO/legal for GDPR/AI Act) which are freely available in open-source intelligence modelling. It is therefore impossible to trace the exact source and link across requirements beyond the direct derivation link contained in the requirements elicitation document in appendix. Further on, the OO goal model reflects core CPS artifacts present in PENTION: `Sampler` → `Sample` → `Analyzer` → `AnalysisResult` → `RiskEvent` → `EvidenceRecord`, with `UseCaseScenario` binding to `Location/Parcel/Vehicle` and `ModelVersion` configuring analysis & explanations. The goal model organizes the requirements into actionable goals and sub-goals suitable for acceptance criteria and KPI design.

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Appendix – Full Requirements List [Authentication Required]

Requirements drawn from trend and topic modelling over EU Drug Detection Projects System Requirements and Enriched via requirements elicitation. Full list here:

<https://tinyurl.com/RequirementsEUNRelatedPention>



Requirements drawn from elicitation, analysis, and formalization of PENTION stakeholders and connected partners is available here:

<https://tinyurl.com/PENTION-Partners-Requirements>

